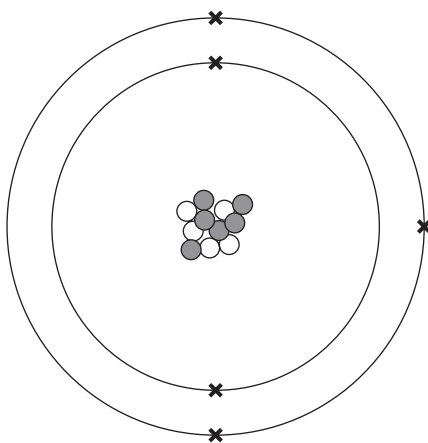


2. (a) Atoms contain particles called electrons, protons and neutrons. The diagram shows an atom of boron.



Complete the sentences.

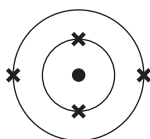
[4]

The nucleus of a boron atom contains five ..... and six .....

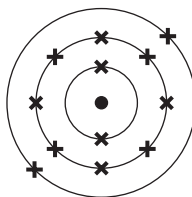
There are five ..... in the shells around the nucleus.

The particles in the atom which have a positive charge are called .....

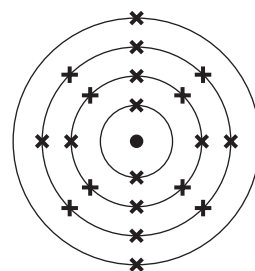
- (b) The diagrams show three different atoms labelled **A**, **B** and **C**. These letters are **not** chemical symbols.



**A**



**B**



**C**

Complete the sentences.

- (i) Atoms **A**, **B** and **C** can be found in Group .....
- (ii) The atomic number of atom **A** is .....
- (iii) The electronic structure of atom **C** is (2, .....).
- (iv) Atom **B** can be found in Period .....

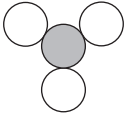
[1]

[1]

[1]

[1]

- (c) The following table shows three substances, their formulae and diagrams that can be used to represent them.

| Substance         | Formula              | Diagram                                                                             |
|-------------------|----------------------|-------------------------------------------------------------------------------------|
| nitrogen trioxide | $\text{NO}_3$        |  |
| methane           | $\text{CH}_4$        |  |
| water             | $\text{H}_2\text{O}$ |  |

- (i) Use the information in the table to work out the key being used to represent the different elements in the diagrams. [2]

 is .....

 is .....

 is .....

 is .....

- (ii) **Using the same key** draw a diagram to represent a molecule of carbon dioxide,  $\text{CO}_2$ . [1]

- (d) The chemical formula of phosphoric acid is  $\text{H}_3\text{PO}_4$ .

(i) State how many phosphorus atoms are present in the formula  $\text{H}_3\text{PO}_4$ . ..... [1]

(ii) Give the **total** number of atoms shown in the formula  $\text{H}_3\text{PO}_4$ . ..... [1]